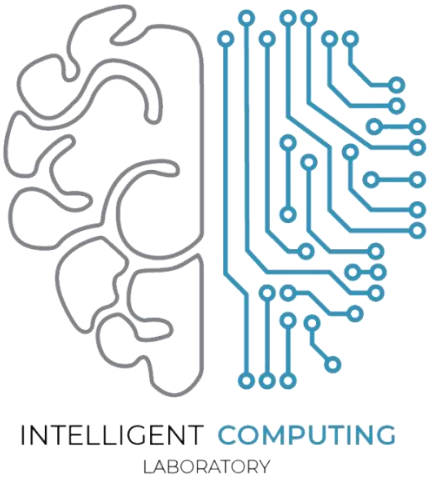
LAPORAN TEORI

PEMBELAJARAN MESIN



NAMA : Anfal Ridho Hanif

NIM : 202431052

KELAS : C

DOSEN : Dr. Efy Yosrita, S. Si., M.Kom

NO.PC : 22

ASISTEN : 1. Muhammad Raka Ilham

2. Sharira Maharani

3. Muhammad Hanif

4. Muhammad Fachrydzi

INSTITUT TEKNOLOGI PLN

TEKNIK INFORMATIKA

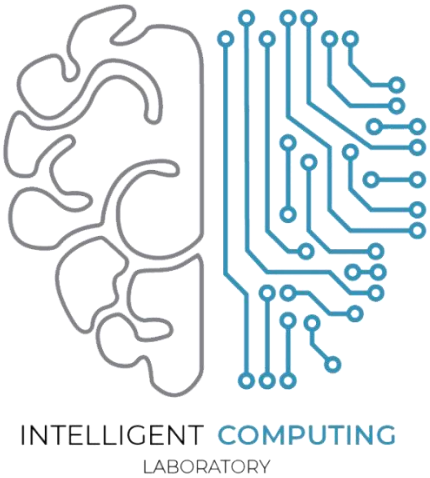
2025/2026

Laporan ke Tanggal Pengumpulan

NIM NAMA

LAPORAN PRAKTIKUM

PEMBELAJARAN MESIN



NAMA : Anfal Ridho Hanif

NIM : 202431052

KELAS : C

DOSEN : Dr. Efy Yosrita, S. Si., M.Kom

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INSTITUT TEKNOLOGI PLN

TEKNIK INFORMATIKA

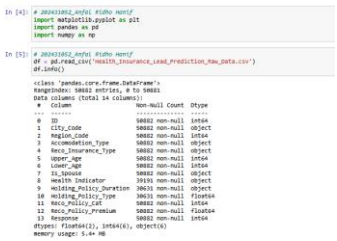
2025/2026

Laporan Praktikum:

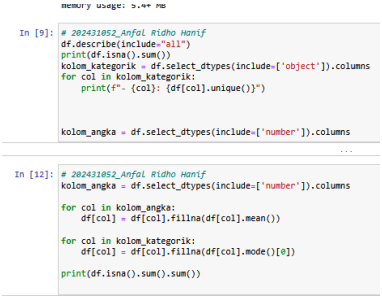
1. EDA (Exploratory Data Analisys):

Dataset: Health_Insurance_Lead_Prediction_Raw_Data adalah sebuah kumpulan calon pelanggan asuransi kesehatan yang digunakan untuk menganalisis karakteristik pelanggan dan memprediksi kemungkinan seseorang tertarik atau tidak menerima produk asuransi kesehatan.

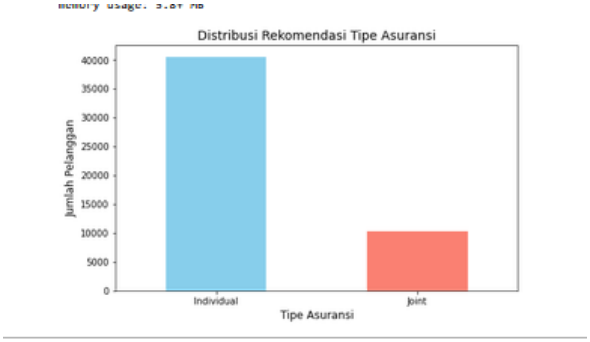
- Mengimport 3 liblary dan info tipe data dataset



- Proses EDA (penanganan missing value, rubah tipe data, dan uniqe text)



- Visualisasi data bersih



2. Scrapping

Website: https://komikcast.info/

Alamat robots.txt: https://komikcast.info/robots.txt

Aturan yang diperbolehkan:

User-agent: *

Allow: /

Disallow: /admin

Disallow: /admin/*

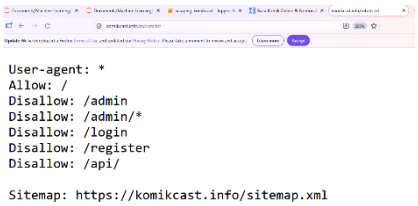
Disallow: /login

Disallow: /register

Disallow: /api/

Sitemap: https://komikcast.info/sitemap.xml

Foto:

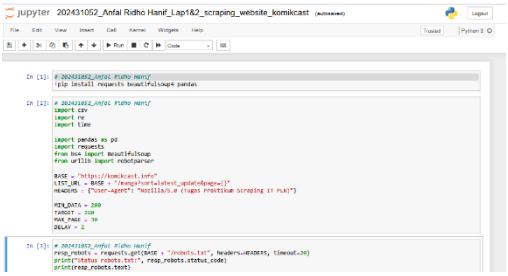


```
User-agent: *
Allow: /
Disallow: /admin
Disallow: /admin/*
Disallow: /login
Disallow: /register
Disallow: /api/

Sitemap: https://komikcast.info/sitemap.xml
```

Langkah ini bertujuan untuk mengetahui aturan dalam melakukan scraping, dari website terdapat larangan yang tidak diizinkan diambil tetapi ada beberapa yang diperbolehkan diambil seperti judul komik, tipe komik, rating, dan chapter.

1. Menimport liblary dan menginstal package untuk menginstal 3 liblary untuk web scraping.



2. Pengecekan aturan untuk web scrapping (yang diizinkan hampir semua kecuali login, admin dan API)



```
In [3]: # 202431052_Anfal Ridho Hanif
resp_robots = requests.get(BASE + "/robots.txt", headers=HEADERS, timeout=20)
print("Status robots.txt:", resp_robots.status_code)
print(resp_robots.text)

rp = robotparser.RobotFileParser()
rp.parse(resp_robots.text.splitlines())
for p in (1, 2):
    url = LIST_URL.format(p)
    print("Boleh diambil?", url, "->", rp.can_fetch("*", url))

Status robots.txt: 200
User-agent: *
Allow: /
Disallow: /admin
Disallow: /admin/*
Disallow: /login
Disallow: /register
Disallow: /api/

Sitemap: https://komikcast.info/sitemap.xml

Boleh diambil? https://komikcast.info/manga?sort=latest_update&page=1 -> True
Boleh diambil? https://komikcast.info/manga?sort=latest_update&page=2 -> True
```

3. Validasi akses web, atribut, dan jumlah blog sebelum dilakukan proses scraping

```
In [4]: # 202431052_Anfal Ridho Hanif
def cari_blok(soup):
    """Cari semua blok data (satu blok = satu kartu komik).

    Tiap kartu punya satu link judul ke /manga/<slug>. Dari link itu kita
    naik ke elemen induk terdekat yang memuat teks 'Rating' = blok kartu.
    """
    blok, sudah = [], set()
    for a in soup.find_all("a", href=re.compile(r"^/manga/[^/?#]+$")):
        if not a.get_text(strip=True):      # Lewati link gambar cover
            continue
        node = a
        for _ in range(6):
            node = node.parent
            if node is None:
                break
            teks = node.get_text(" ", strip=True)
            if "Rating" in teks and "Status" in teks:
                break
        else:
            node = None
        if node is not None and id(node) not in sudah:
            sudah.add(id(node))
            blok.append(node)
    return blok


def ambil_data(b):
    """Ambil tiap atribut dari satu blok (hasilnya masih teks)."""
    judul = b.find("a", href=re.compile(r"^/manga/[^/?#]+$"), string=True)
    judul = judul.get_text(strip=True) if judul else ""
```

4. Proses scraping data dari web dengan pagination

```
In [5]: # 202431052_Anfal Ridho Hanif
result = []
halaman = 1
while halaman <= MAX_PAGE:
    url = LIST_URL.format(halaman)
    r = requests.get(url, headers=HEADERS, timeout=20)
    print(f"[Halaman {halaman}] status_code = {r.status_code}")
    if r.status_code != 200:
        print("Status bukan 200, berhenti.")
        break

    soup = BeautifulSoup(r.text, "html.parser")
    blok = cari_blok(soup)
    print(f"[Halaman {halaman}] jumlah blok data = {len(blok)}")
    if not blok:
        break

    for b in blok:
        result.append(ambil_data(b))

    if sum(valid(d) for d in result) >= TARGET:
        break
    halaman += 1
    time.sleep(DELAY)

print("\nTotal blok terkumpul:", len(result))
```

```
[Halaman 1] status_code = 200
[Halaman 1] jumlah blok data = 24
[Halaman 2] status_code = 200
[Halaman 2] jumlah blok data = 24
```

5. Hasil scraping

```
Total blok terkumpul: 408

In [6]: # 202431052_Anfal Ridho Hanif
for i, d in enumerate(result, start=1):
    print(i, d["judul"], "|", d["tipe"], "|", d["status"],
          "|", d["chapter_terbaru"], "|", d["rating"])
```

```
206 Kichiku Eiyuu | Manga | Ongoing | 119.0 | 9.7
207 SSS-Class Suicide Hunter | Manhwa | Ongoing | 158.0 | 9.5
208 Misutereareta Trap Master no Nariagari | Manga | Ongoing | 8.0 | 0.0
209 Yami Sute Ana | Manga | Ongoing | 2.0 | 0.0
210 Immortal's Way Of Life | Manhwa | Ongoing | 36.0 | 10.0
211 Barbarian's Adventure In A Fantasy World | Manhwa | Ongoing | 74.0 | 10.0
212 The Hated Black Mage Returns | Manga | Ongoing | 28.0 | 7.3
213 The Zero Chronicles of the Nine Dreams | Manhwa | Ongoing | 59.0 | 0.0
214 Since My Previous Life Was A Wise Man I Can Afford To Live | Manga | Ongoing | 84.0 | 8.0
215 Now It's Your Turn To Regret | Manhwa | Ongoing | 20.0 | 0.0
216 Lord of Summons! Sudden Mutation | Manhua | Ongoing | 564.0 | 0.0
217 A Wicked Husband | Manhwa | Ongoing | 51.0 | 0.0
218 Defending the Villain's Innocence | Manhwa | Ongoing | 15.0 | 10.0
219 Against the Gods | Manhua | Ongoing | 7199.0 | 8.0
220 Hatsukoi Arugorizumu | Manga | Ongoing | 7.0 | 0.0
221 I'm Really Not A Demon Beast | Manhua | Ongoing | 14.0 | 8.0
222 1st in Class Hides Regression | Manhwa | Ongoing | 26.0 | 8.0
223 My Academic System Is Making Me Date?! | Manhua | Ongoing | 39.0 | 10.0
224 Star Ginseng Store | Manhwa | Ongoing | 220.0 | 10.0
```

3. Regresi Linear (Menggunakan Hasil Scrapting)

- Langkah 1: mengimport liblary dan dataset hasil scraping

```
In [1]: # 202431052_Anfal Ridho Hanif
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

In [2]: # 202431052_Anfal Ridho Hanif
df = pd.read_csv("komikcast_data.csv")
display(df.head())
print(df.shape)
print(df.dtypes)
```

	judul	tipe	status	chapter_terbaru	rating
0	Reincarnated as the Archduke's Son	Manhwa	Ongoing	56.0	9.7
1	The Supreme Demon Swordmaster	Manhua	Ongoing	60.0	10.0
2	Black Killer Whale Baby	Manhwa	Ongoing	128.0	10.0
3	Student Jo-Eun	Manhwa	Ongoing	95.0	10.0
4	The Saint Levels Up Through Necromancy	Manhwa	Ongoing	45.0	9.0

```
(266, 5)
judul              object
tipe               object
status             object
chapter_terbaru    float64
rating             float64
```

- Langkah 2: Membuat dataset kosong, menginisialisasi fitur (x) dan target (y), menggunakan data latih dan data uji, dan prediksi.

```
In [3]: # 202431052_Anfal Ridho Hanif
        df = df.dropna(subset=["chapter_terbaru", "rating"])
        print("Jumlah data setelah dropna:", len(df))

        Jumlah data setelah dropna: 266

In [4]: # 202431052_Anfal Ridho Hanif
        x = df[["chapter_terbaru"]]   # fitur (2 dimensi)
        y = df["rating"]              # target

In [5]: # 202431052_Anfal Ridho Hanif
        x_train, x_test, y_train, y_test = train_test_split(
            x, y, test_size=0.2, random_state=0
        )
        print("Data latih:", x_train.shape[0])
        print("Data uji  :", x_test.shape[0])

        Data latih: 212
        Data uji  : 54
```

- Pembuatan model, evaluasi, keofisen, Intercept, MSE, dan R2 Score

```
In [12]: # 202431052_Anfal Ridho Hanif
         model = LinearRegression()
         model.fit(x_train, y_train)
         y_pred = model.predict(x_test)

         hasil = pd.DataFrame({
             "chapter_terbaru": x_test["chapter_terbaru"].values,
             "rating_aktual": y_test.values,
             "rating_prediksi": y_pred,
         })
         display(hasil.head(10))

In [8]: # 202431052_Anfal Ridho Hanif
        koef = model.coef_
        intercept = model.intercept_
        mse = mean_squared_error(y_test, y_pred)
        r2 = r2_score(y_test, y_pred)

        print("Koefisien :", koef)
        print("Intercept :", intercept)
        print("MSE       :", mse)
        print("R2 Score  :", r2)

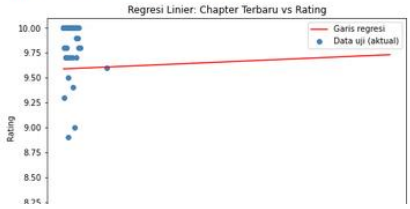
        Koefisien : [1.98958881e-05]
        Intercept : 9.588007319935606
        MSE       : 0.3219241938914124
        R2 Score  : -0.08830800105193659
```

- Visualisasi dengan membuat grafik regresi linear yang membandingkan nilai rating_aktual dengan hasil prediksi berdasarkan chapter terbaru

```
In [11]: # 202431052_Anfal Ridho Hanif
         a = intercept
         b = koef[0]
         print(f"Rating = {a:.4f} + {b:.6f} * chapter_terbaru")

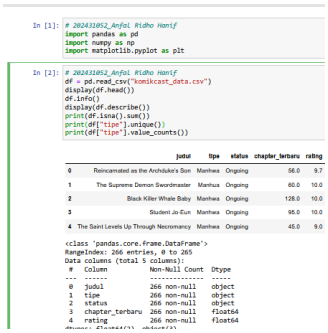
         plt.figure(figsize=(8, 5))
         plt.scatter(x_test, y_test, color="steelblue", label="Data uji (aktual)")
         plt.plot(x_test.sort_values("chapter_terbaru"),
                  model.predict(x_test.sort_values("chapter_terbaru")),
                  color="red", label="Garis regresi")
         plt.xlabel("Chapter terbaru")
         plt.ylabel("Rating")
         plt.title("Regresi Linier: Chapter Terbaru vs Rating")
         plt.legend()
         plt.show()

         Rating = 9.5880 + 0.000020 * chapter_terbaru
```



4. Decision Tree (Hasil Scraping)

- Langkah 1, import liblary pandas, numpy, dan matplolib untuk kebutuhan proses pengolahan serta visualisasi data, dataset hasil scraping digunakan dengan liblary pandas dan menampilkan informasi dataset



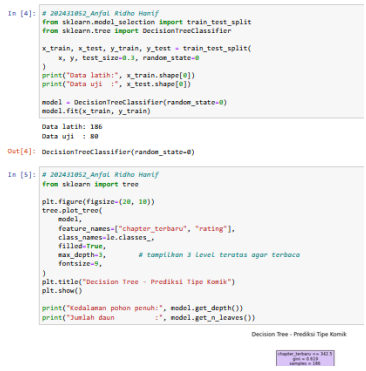
```
In [1]: # 202431052_Anfal Ridho Hanif
        import pandas as pd
        import numpy as np
        import matplotlib.pyplot as plt

In [2]: # 202431052_Anfal Ridho Hanif
        df = pd.read_csv("komikcast_data.csv")
        display(df.head())
        df.info()
        display(df.describe())
        print(df.isna().sum())
        print(df["tipe"].unique())
        print(df["tipe"].value_counts())
```

	judul	tipe	status	chapter_terbaru	rating
0	Reincarnated as the Archduke's Son	Manhwa	Ongoing	56.0	9.7
1	The Supreme Demon Swordmaster	Manhua	Ongoing	60.0	10.0
2	Black Killer Whale Baby	Manhwa	Ongoing	128.0	10.0
3	Student Jo Eun	Manhwa	Ongoing	95.0	10.0
4	The Saint Levels Up Through Necromancy	Manhwa	Ongoing	45.0	9.0

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 266 entries, 0 to 265
Data columns (total 5 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   judul            266 non-null    object
 1   tipe             266 non-null    object
 2   status           266 non-null    object
 3   chapter_terbaru  266 non-null    float64
 4   rating           266 non-null    float64
```

- Langkah 2, Pembuatan model yang akan digunakan untuk menguji 186 data latih dan 80 data uji. Menggunakan atribut chapter_terbaru dan rating untuk memprediksi tipe komik



```
In [4]: # 202431052_Anfal Ridho Hanif
        from sklearn.model_selection import train_test_split
        from sklearn.tree import DecisionTreeClassifier

        x_train, x_test, y_train, y_test = train_test_split(
            x, y, test_size=0.3, random_state=0
        )
        print("Data latih:", x_train.shape[0])
        print("Data uji  :", x_test.shape[0])

        model = DecisionTreeClassifier(random_state=0)
        model.fit(x_train, y_train)

        Data latih: 186
        Data uji  : 80

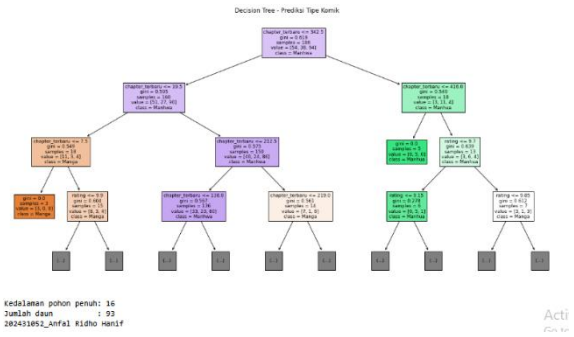
Out[4]: DecisionTreeClassifier(random_state=0)

In [5]: # 202431052_Anfal Ridho Hanif
        from sklearn import tree

        plt.figure(figsize=(20, 10))
        tree.plot_tree(
            model,
            feature_names=["chapter_terbaru", "rating"],
            class_names=le.classes_,
            filled=True,
            max_depth=3,        # tampilkan 3 level teratas agar terbaca
            fontsize=9,
        )
        plt.title("Decision Tree - Prediksi Tipe Komik")
        plt.show()

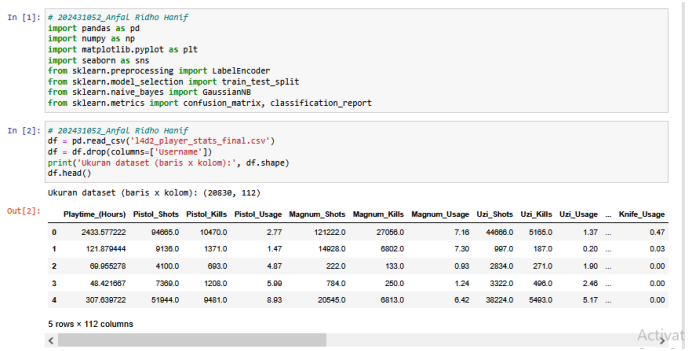
        print("Kedalaman pohon penuh:", model.get_depth())
        print("Jumlah daun          :", model.get_n_leaves())
```

- Langkah 3, hasil evaluasi dari model decision tree berbentuk pohon klasifikasi untuk memprediksi tipe komik berdasarkan 2 atribut tadi dengan kedalaman pohon maksimum 16 tingkat dan 93 daun. (Kategori: Manhwa, Manhua, dan Manga).



5. Naive Bayes

- Langkah 1, Import Liblary dan beberapa fungsi dari sklearn untuk proses klasifikasi, yang kemudian membaca dataset csv yang digunakan dan menghapus kolom username dan ditampilkan ukuran dataset dan lima baris pertama untuk melihat struktur datanya.



```
In [1]: # 202431052_Anfal Ridho Hanif
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import confusion_matrix, classification_report

In [2]: # 202431052_Anfal Ridho Hanif
df = pd.read_csv('1442_player_stats_final.csv')
df = df.drop(columns=['Username'])
print('Ukuran dataset (baris x kolom):', df.shape)
df.head()

Ukuran dataset (baris x kolom): (20830, 112)
```

Out[2]:

	Playtime_(Hours)	Pistol_Shots	Pistol_Kills	Pistol_Usage	Magnum_Shots	Magnum_Kills	Magnum_Usage	Uzi_Shots	Uzi_Kills	Uzi_Usage	...	Knife_Usage
0	2433.577222	94665.0	10470.0	2.77	121222.0	27056.0	7.16	44666.0	5165.0	1.37	...	0.47
1	121.879444	9136.0	1371.0	1.47	14928.0	6802.0	7.30	997.0	187.0	0.20	...	0.03
2	69.955278	4100.0	693.0	4.87	222.0	133.0	0.93	2834.0	271.0	1.90	...	0.00
3	48.421667	7369.0	1208.0	5.99	784.0	250.0	1.24	3322.0	496.0	2.46	...	0.00
4	307.639722	51944.0	9481.0	8.93	20545.0	6813.0	6.42	38224.0	5493.0	5.17	...	0.00

5 rows × 112 columns

- Langkah 2, memeriksa struktur dan statistik dataset menggunakan info() dan describe(), dan memisahan atribut bertipe numerik dan kategorikal berdasarkan tipe datanya sehingga hasilnya menunjukan bahwa atribut kategorikalnya ditemukan ”Difficulty”, dengan empat kategori (Normal, Expert, Advanced, dan Easy).



```
In [3]: # 202431052_Anfal Ridho Hanif
df.info()

In [4]: # 202431052_Anfal Ridho Hanif
df.describe()

In [5]: # 202431052_Anfal Ridho Hanif
num_cols = df.select_dtypes(include=[np.number]).columns.tolist()
print('Jumlah fitur numerik:', len(num_cols))
print('Daftar fitur numerik:', num_cols)

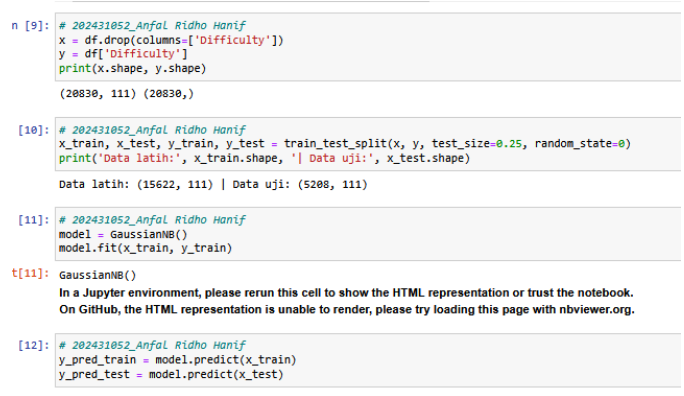
In [6]: # 202431052_Anfal Ridho Hanif
cat_cols = df.select_dtypes(exclude=[np.number]).columns.tolist()
print('Jumlah fitur kategorikal:', len(cat_cols))
print('Daftar fitur kategorikal:', cat_cols)

Jumlah fitur kategorikal: 1
Daftar fitur kategorikal: ['Difficulty']

In [7]: # 202431052_Anfal Ridho Hanif
for col in cat_cols:
    print(f"Nilai unik '{col}':", df[col].unique())

Nilai unik 'Difficulty': <StringArray>
['Normal', 'Expert', 'Advanced', 'Easy']
Length: 4, dtype: str
```

- Langkah 3, Memisahkan fitur dan label target dengan menghapus kolom difficulty dari dataset, kemudian membagi data menjadi data latih dan data uji dengan rasio 75:25 selanjutnya, model Gaussian Naive Bayes dibuat , dilatih menggunakan data latih, dan digaunakan untuk melakukan prediksi terhadap data latih maupun data uji.



```
n [9]: # 202431052_Anfal Ridho Hanif
x = df.drop(columns=['Difficulty'])
y = df['Difficulty']
print(x.shape, y.shape)

(20830, 111) (20830,)

[10]: # 202431052_Anfal Ridho Hanif
x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.25, random_state=0)
print('Data latih:', x_train.shape, '| Data uji:', x_test.shape)

Data latih: (15622, 111) | Data uji: (5208, 111)

[11]: # 202431052_Anfal Ridho Hanif
model = GaussianNB()
model.fit(x_train, y_train)

t[11]: GaussianNB()
In a Jupyter environment, please rerun this cell to show the HTML representation or trust the notebook.
On GitHub, the HTML representation is unable to render, please try loading this page with nbviewer.org.

[12]: # 202431052_Anfal Ridho Hanif
y_pred_train = model.predict(x_train)
y_pred_test = model.predict(x_test)
```

- Mengevaluasi hasil prediksi model pada data latih menggnunakan confusionmatrix dan classification report, yang menampilkan nilai precision, recall, f1-score, serta accuracy. Yang kemudia membuat hasil visualisasinya dari confusion matrix dalam bentuk heatmap untuk membandingkan hasil prediksi dengan label sebenarnya.

```
In [13]: # 202431052_Anfal Ridho Hanif
labels = le.classes_
cm_train = confusion_matrix(y_train, y_pred_train)
print('===== Evaluasi Data Latih (Training) =====')
print('Confusion Matrix:\n', cm_train)
print('\nClassification Report:\n', classification_report(y_train, y_pred_train, target_names=labels, zero_division=0))

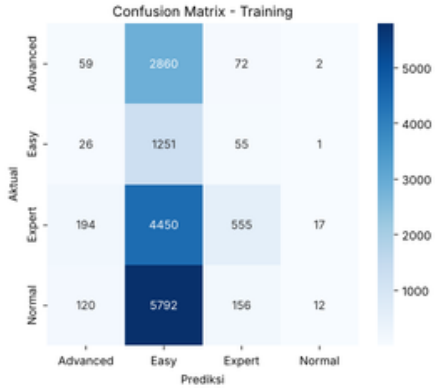
plt.figure(figsize=(6, 5))
sns.heatmap(cm_train, annot=True, fmt='d', cmap='Blues', xticklabels=labels, yticklabels=labels)
plt.title('Confusion Matrix - Training')
plt.xlabel('Prediksi')
plt.ylabel('Aktual')
plt.show()
```

```
===== Evaluasi Data Latih (Training) =====
Confusion Matrix:
 [[  59 2860   72    2]
 [  26 1251   55    1]
 [ 194 4450  555   17]
 [ 120 5792  156   12]]

Classification Report:
              precision    recall  f1-score   support

    Advanced       0.15      0.02      0.03      2993
        Easy       0.09      0.94      0.16      1333
      Expert       0.66      0.11      0.18      5216
      Normal       0.38      0.00      0.00      6080

    accuracy                           0.12     15622
   macro avg       0.32      0.27      0.10     15622
weighted avg       0.40      0.12      0.08     15622
```



- Mengevaluasi hasil prediksi model Gaussian Naive Bayes pada data uji dengan menghitung confusion matrix danclassification report yang menampilkan nilai precision, recall, f1-score, serta accuracy. Membuat visualisasi confusion matrix dalam bentuk heatmap untuk membandingkan hasil prediksi model dengan label aktual pada data uji.

```
In [14]: # 202431052_Anfal Ridho Hanif
cm_test = confusion_matrix(y_test, y_pred_test)
print('===== Evaluasi Data Uji (Testing) =====')
print('Confusion Matrix:\n', cm_test)
print('\nClassification Report:\n', classification_report(y_test, y_pred_test, target_names=labels, zero_division=0))

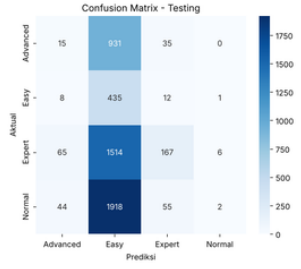
plt.figure(figsize=(6, 5))
sns.heatmap(cm_test, annot=True, fmt='d', cmap='Blues', xticklabels=labels, yticklabels=labels)
plt.title('Confusion Matrix - Testing')
plt.xlabel('Prediksi')
plt.ylabel('Aktual')
plt.show()
```

```
===== Evaluasi Data Uji (Testing) =====
Confusion Matrix:
 [[ 15  931   35    0]
 [  8  435   12    1]
 [ 65 1514  167    6]
 [ 44 1918   55    2]]

Classification Report:
              precision    recall  f1-score   support

    Advanced       0.11      0.02      0.03       981
        Easy       0.09      0.95      0.17       456
      Expert       0.62      0.10      0.17      1752
      Normal       0.22      0.00      0.00      2019

    accuracy                           0.12      5208
   macro avg       0.26      0.27      0.09      5208
weighted avg       0.32      0.12      0.08      5208
```



- Membuat data baru berdasarkan dari seluruh nilai median fitur yang kemudian akan diguanakan untuk mengatur fitur Playtime_(Hours) menjadi 500 dan Average_Friendly_Fire menjadi 40. Selanjutnya, model Gaussian Navie Bayes melakukan prediksi tingkat kesulitan, kemudia hasil prediksi ditampilkan dalam bentuk kode prediksi dan nama kategori Difficulty

In [15]:
```
# 202431052_Anfal Ridho Hanif
data_baru = x.median().to_frame().T  # profil pemain tipikal (median tiap fitur)
data_baru['Playtime_(Hours)'] = 500
data_baru['Average_Friendly_Fire'] = 40
pred_baru = model.predict(data_baru)
print('Kode prediksi :', pred_baru[0])
print('Difficulty    :', le.inverse_transform(pred_baru)[0])
```

```
Kode prediksi : 1
Difficulty    : Easy
```